Jame's #CS2 crashes at 11-11 | PGL Major (HEARTBREAKING)
MacMate
MAJOR
CRASH
Watch on YouTube

Jame on crucial crash: "We had to accept it as is and move on" (subtitles)
HLTVorg
HLTV
Jame on the crash and loss vs. G2
PGL CS2 Major Copenhagen
Watch on YouTube

TBZ

SEMESTERARBEIT 5.1

RS:GO

Memory-Safe Gaming mit Rust

Speaker

DAVID UNTERGUGGENBERGER

Date

15 APRIL 2026

PROBLEM & ZIEL

PROBLEM

Software Crashes
Security Vulnerabilities

MOTIVATION

Memory-Safe Gaming
Game Mechanic Verstehen
Von Scratch

VISION

Shooter in Rust schreiben
Ohne Game Engine

DISCLAIMER

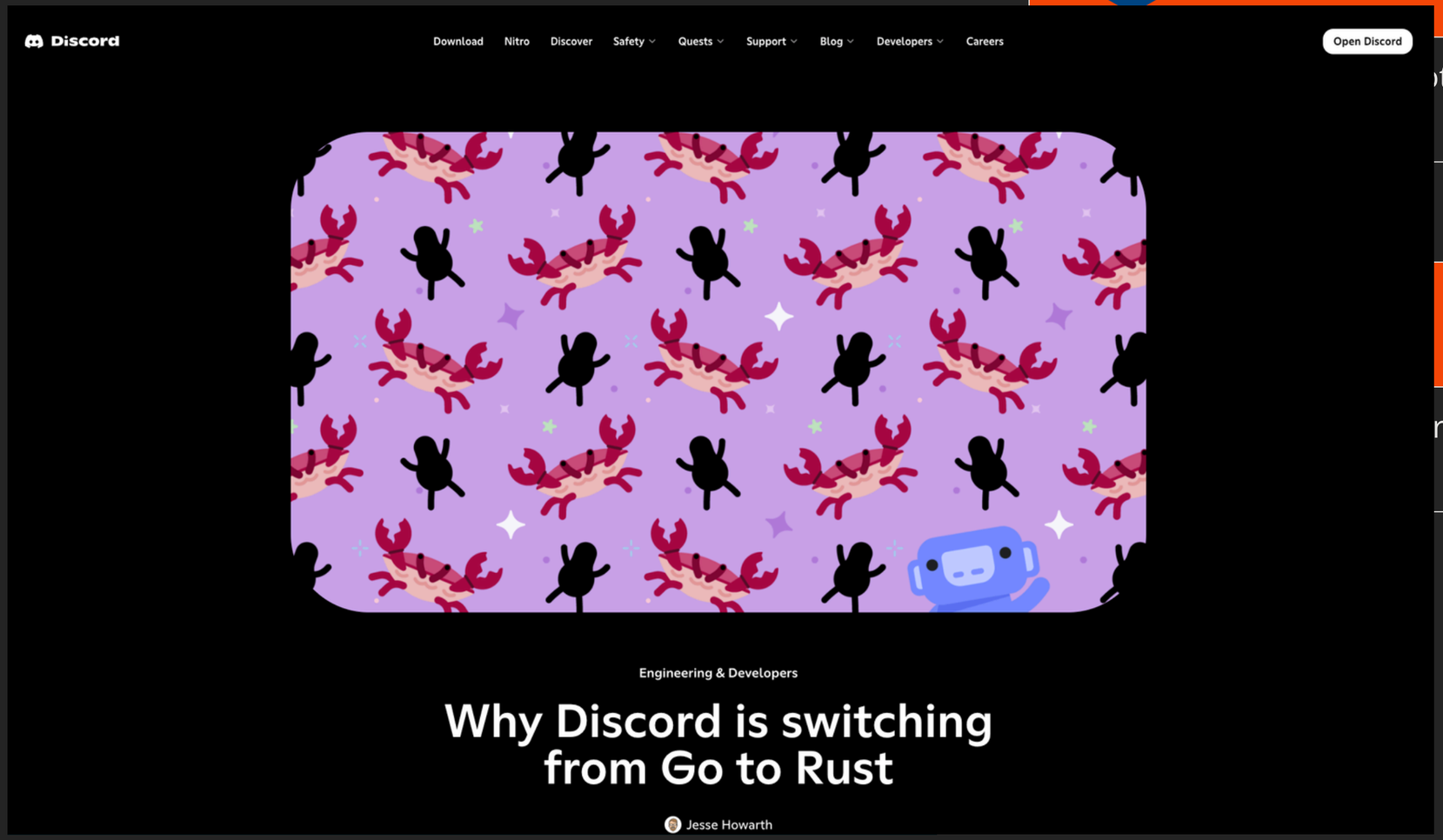
Kein AAA-Studio

WARUM RUST?

WARUM RUST?

PROBLEME	Memory Leak: Speicher nicht aufgeräumt
Segfault: Zugriff auf verbotenen Speicher	Race Condition: Gleichzeitige Zugriffe = Chaos

WARUM RUST?



PROBLEME	
	Memory Leak: Speicher nicht aufgeräumt
tenen	Race Condition: Gleichzeitige Zugriffe = Chaos

	Memory Safety: Borrow Checker
r, aber	Zero Crashes: "If it compiles, it runs forever"

RSGO ARCHITEKTUR

FRONTEND

three.js

BACKEND

Rust

KOMMUNIKATION

WebSocket

DEPLOYMENT

RSGO ARCHITEKTUR

FRONTEND

three.js

BACKEND

Rust

KOMMUNIKATION

WebSocket

DEPLOYMENT

docker

FEATURES

CORE GAMEPLAY

Movement: WASD + Mouse
Perspektive: First-Person View
Physics: 3D Grafik und Kollision

COMBAT SYSTEM

Waffe: Revolver mit Decal
Shooting: Recoil-Animation
Hit-Detection: Server-side Validation

MULTIPLAYER FEATURES

Players: Mehrere gleichzeitig
Kill-Feed: Real-time Notifications
UI HUD: Map und Kompass

DEMO

RSGO.UNTERGUGGENBERGER.CH

MESSAGE

FAZIT & AUSBLICK

“Often I was scared and uncertain whether I could make it because I thought it was too hard. But over time I realized: It was never hard, it was just new."

-Elon Musk

ERFOLGE

- Shooter Game von Scratch gebaut
- Game Mechanic Verständnis
- WebSockets schnell genug für FPS-Gaming
- Mehrere Spieler gleichzeitig ohne Lag

AUSBLICK

- Weitere Waffen
- Neue Maps
- Neue Modi (CS:GO Style Plant/Defuse Mechanic)
- Ranked System
- Monetarisierung (Cosmetics, Waffen Skins, Battle Pass)

DANKE

FRAGEN?

https://github.com/JumpiiX/rsgo-game-client
https://github.com/JumpiiX/rsgo-frontend